Rel-Sortability of Time-Unrolled Causal Graphs: Exact Values, Finite-Sample Saturation, and Pre-Registered Real-Data Tests

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Abstract

Simulated causal-discovery benchmarks can be solved by sorting variables by marginal statistics (var-, R^2- and rel-sortability). Time-unrolled DAGs have recently been proposed as a remedy, with a sufficient condition for their rel-sortability to tend to $1/2$. We give the exact rel-sortability of the time-unrolled DAG for every summary graph, every horizon T and every edge, with an explicit $O(1/T)$ rate. The condition of the earlier result is also necessary, and every non-strongly-connected summary graph, for instance any river network, stays strictly above $1/2$. This excess is caused by the fixed start at $t = 1$: with a stationary start the value is exact for every T and depends only on the source strongly connected components of the summary graph. We show that block-mean aggregation and point subsampling leave this value unchanged but move var- and R^2-sortability to different limits. A finite-sample analysis explains why the population value is hard to see in real series: raw correlation tests are inconsistent under autocorrelation, a latent common driver saturates the relatives count at $n \approx z_c^2/\ell^2$, and proxy conditioning works only with near-perfect reliability and a rank-one driver. All graph-theoretic results and the algebraic core of the finite-sample results are machine-checked in Lean 4 (116 theorems). In pre-registered real-data studies, a river benchmark fails as the theory predicts, and a physical testbed with designed independent sources gives a partial positive result, including out-of-sample predictions of detection rates. A fourth pre-registration (E1) asks whether the theorem's edge values beat eight naive predictors on a multi-layer light-tunnel series. It does, mostly through predicted ties, and only for the linearly dependent relatives (on the published graph one naive predictor is significantly better on the strictly ordered edges). A lagged-relatives estimator (E2) raises detection on wind-tunnel data, but its pre-registered graded criterion fails, as P9 did.

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1. Introduction

Simulated benchmarks drive much of causal discovery, and several of their regularities can be exploited without any causal reasoning. Marginal variances tend to grow along the causal order (*var-sortability*; Reisach et al., 2021), so a method that sorts by variance does well on data from common generators, and it explains the strong benchmark performance of some continuous-optimisation methods, a family that includes NOTEARS (Zheng et al., 2018). The coefficient of determination R^2 has the same property and is scale invariant (Reisach et al., 2023); internally standardised structural models are not var-sortable by construction and, empirically, mostly not R^2-sortable (Ormaniec et al., 2025); and Herman et al. (2025) relate the R^2 pattern to colliders and confounders and propose a generator that avoids it. Most recently, Reisach et al. (2026) defined *rel-sortability*, which orders a node by the number of its *relatives* (nodes joined to it by an open path, marginally dependent on it under faithfulness). Relatives are a property of the graph alone, so the criterion cannot be removed by reweighting. For random DAGs it is high; for *time-unrolled* DAGs built from a summary graph with self-loops, Reisach et al. (2026, Thm. 4) show that it tends to $1/2$ if every weakly connected component of the summary graph is strongly connected, and they propose such graphs as benchmarks.

Three questions remain open, and they matter for real time series, whose summary graphs are often feed-forward (rivers, supply chains, signalling cascades) and whose observations are aggregated or subsampled: (Q1) what is the rel-sortability of a time-unrolled DAG when the summary graph is not strongly connected; (Q2) does it depend on how the series is started and observed; and (Q3) can a relatives count be estimated from one finite, autocorrelated record, so that a method could actually exploit it?

Contributions. (1) The exact rel-sortability of every edge

of the time-unrolled DAG G_T, for every summary graph and every T (Theorem 1), its limit $\frac{1}{2} + |E_\times|/(2|E_\mathrm{S}|)$ with $E_\times$ the edges between strongly connected components, and an explicit $O(1/T)$ rate (Theorem 2). The condition of Reisach et al. (2026, Thm. 4) is also necessary (Corollary 3). The excess above $1/2$ for, e.g., chains and out-trees is a boundary effect of the deterministic start at $t = 1$ (Proposition 4). **(2)** A stationary version (latent burn-in of $p - 1$ slices or an infinite past) whose value is exact for every T and is fixed by the *source* strongly connected components (Theorem 5); river networks exceed $1/2$ iff they contain a confluence (Corollary 6). **(3)** Under temporal aggregation the relatives are unchanged, whereas var- and R^2-sortability converge to those of a long-run structural model under block means and stay at the snapshot values under subsampling (Section 4). **(4)** Finite-sample theory (F1–F3, Section 5): a recovery bound with an effective sample size and the inconsistency of the nominal-n test under autocorrelation; saturation by a latent driver at $n \approx z_c^2/\ell^2$; and a proxy-conditioning analysis showing that the proxy needs reliability close to 1 and the driver must be one-dimensional. **(5)** Machine-checked proofs in Lean 4 with Mathlib of all graph results and of the deterministic core of F1–F3 (116 theorems, standard axioms only; Section 6). Every theorem below names its Lean counterpart. **(6)** Four pre-registered real-data studies (Section 7). On CausalRivers (Stein et al., 2025) the relatives sort carries no orientation information, and F2/F3 predict the observed dependence rates. On the Causal Chambers (Gamella et al., 2025) two of three pre-registered wind-tunnel criteria pass, a light-tunnel replication passes all three, and F1/F2 predict held-out detection rates within 0.02. A fourth study (E1) tests the theorem against eight naive edge predictors on a light-tunnel series with multi-layer ground truth: it passes 8/8, but under a refined graph of linearly dependent relatives and mostly through predicted ties, and it fails for the published graph on strictly ordered edges. A lagged-relatives estimator (E2) passes three of four criteria and fails the graded one. We report every failed pre-registered test in the main text (Table 2).

The graph results of contributions (1)–(2) are short consequences of a walk-length lemma and of Reisach et al. (2026, Lemma 1). Their value is that they settle the question completely, are machine-checked, and change what a benchmark designer should conclude: much of the rel-sortability of time-unrolled benchmarks is an artefact of how the simulation is started. The finite-sample results use textbook tools; their value is that they predicted, quantitatively and in one case out of sample, the outcome of real-data tests.

Related work. Beyond the sortability papers above, Lohse & Wahl (2025) measure var- and R^2-sortability of real and simulated time series and note that cyclic pairs push the signal towards $1/2$; they give no theorem on relatives or aggregation. Identifiability from subsampled (Gong et al., 2015) and aggregated (Gong et al., 2017) series is a separate question from sortability. Time-series discovery methods such as PCMCI (Runge et al., 2019) test conditional independences, and our finite-sample results concern only the marginal tests that define relatives.

2. Setting

For a directed graph, $\mathrm{anc}(y)$ and $\mathrm{desc}(y)$ contain y; the relatives of y are $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{anc}(y)) = \{z : \mathrm{anc}(y) \cap \mathrm{anc}(z) \neq \emptyset\}$ and $\rho(y) = |\mathrm{rel}(y)|$. For an edge (x, y), $s_\rho(x, y) = 1, \frac{1}{2}, 0$ if $\rho(x) <, =, > \rho(y)$, and $s_\rho(G)$ is the mean over edges (Reisach et al., 2026). A value $1/2$ means the criterion carries no orientation information.

A *summary graph* $G_\mathrm{S} = (V, E_\mathrm{S})$ has $p \geq 2$ nodes and a self-loop at every node. The *time-unrolled DAG* G_T has nodes $v(t)$, $t \in \{1, \ldots, T\}$, and an edge $(i(t), j(t{+}1))$ for each $(i, j) \in E_\mathrm{S}$ and $t < T$, so every summary edge has $T - 1$ copies. $d(u, v)$ is the shortest-path length in G_S (self-loops ignored; $d(v, v) = 0$; ∞ if unreachable), and anc_S, rel_S are taken in G_S. SCC means strongly connected component. $E_\times \subseteq E_\mathrm{S}$ is the set of edges between different SCCs and $E_\circ = E_\mathrm{S} \setminus E_\times$. A *source* SCC has no incoming edge from another SCC, and $\mathrm{Src}(v)$ is the set of source SCCs that reach v.

3. Exact Rel-Sortability of Time-Unrolled DAGs

Theorem 1 (every edge, every T)**.** *For $(i, j) \in E_\mathrm{S}$ and $1 \leq t < T$, the edge $(i(t), j(t{+}1))$ of G_T has $s_\rho = 1$ if some $u \in V$ satisfies $d(u, j) \leq t \leq d(u, i)$, and $s_\rho = 1/2$ otherwise. The value does not depend on T.*

Lean: `edge_score_dist, edge_score, strictCond_iff_dist`.

Sketch. Every edge of G_T advances time by one, so $u(s)$ is an ancestor of $v(t)$ iff $d(u, v) \leq t - s$ (pad a shortest path with self-loops). Hence the roots of $v(t)$ are $\{u(1) : d(u, v) \leq t - 1\}$. Since $i(t)$ is a parent of $j(t{+}1)$, $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$, and the inclusion is strict iff $j(t{+}1)$ has a root that $i(t)$ lacks (Reisach et al., 2026, Thm. 1, Lemma 1). Full proof in Appendix A. (The Lean proof characterises rel in G_T directly and does not use their Lemma 1.)

Theorem 2 (exact limit and rate)**.** *For every $T \geq 2$, with $L(G_\mathrm{S}) = \frac{1}{2} + \frac{|E_\times|}{2|E_\mathrm{S}|}$,*

$$0 \leq s_\rho(G_T) - L(G_\mathrm{S}) \leq \frac{\min(p{-}1, T{-}1)\,|E_\circ|}{2(T-1)|E_\mathrm{S}|} \leq \frac{p-1}{2(T-1)},$$

so $\lim_{T\to\infty} s_\rho(G_T) = L(G_\mathrm{S})$.

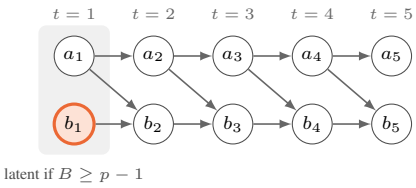


Figure 1. Summary chain $a \to b$ unrolled to $T = 5$. Every a_t has the relatives $\{a_s\} \cup \{b_s : s \geq 2\}$ ($2T - 1$ nodes); every b_t with $t \geq 2$ also has b_1 ($2T$ nodes). So each edge $a_t \to b_{t+1}$ is strictly sorted (finite limit $L = 2/3$), but only because of the root b_1 at the fixed start. With a latent burn-in slice, both counts are $2T$ and $s_\rho^W = 1/2$.

Lean: `sRho_unrolled_exact`, `sRho_unrolled_bounds`, `tendsto_sRho_GT`.

Sketch. For a crossing edge, j does not reach i; take $u = j$ in Theorem 1: all $T - 1$ copies score 1. For an edge inside an SCC, i and j are reached by the same nodes, which forces $t \leq d(u, i) \leq p - 1$, so at most $p - 1$ copies score 1.

Corollary 3 (necessary and sufficient condition)**.** $\lim_T s_\rho(G_T) = 1/2$ *iff every weakly connected component of* G_S *is strongly connected. Otherwise* $s_\rho(G_T) \geq \frac{1}{2} + \frac{1}{2|E_S|}$ *for every* T*. The "if" half is Reisach et al. (2026, Thm. 4). If* G_S *is acyclic apart from self-loops, with* m *other edges,* $L = \frac{1}{2} + \frac{m}{2(m+p)}$ *whatever its shape (chain, in-tree, out-tree, collider).*

Lean: `tendsto_half_iff`, `crossEdges_eq_empty_tfae`, `sRho_unrolled_ge_of_cross`, `reisach_theorem4`, `limitValue_dag`.

Proposition 4 (the excess is a boundary effect)**.** *For* $t \geq p$, $\rho(v(t)) = T\,|\mathrm{rel}_S(v)| - \sum_{w \in \mathrm{rel}_S(v)} d(\mathrm{anc}_S(v), w)$.

Lean: `rho_unrolled_large_t`, `rho_unrolled_eq`.

So for a crossing edge with $\mathrm{rel}_S(i) = \mathrm{rel}_S(j)$ (root to child in an out-tree, a chain), the strict order rests on $O(1)$ extra relatives out of $\Theta(T)$, all located in the first p slices. Figure 1 shows the smallest case. A sample-based relatives count on a long record cannot see such a difference, which motivates the stationary version.

Theorem 5 (stationary start)**.** *Observe the window of the last* $T \geq 2$ *slices of* G_{B+T} *with a latent burn-in* $B \geq p - 1$ *(or of the infinite-past graph on* $V \times \mathbb{Z}$*), with relatives computed in the full graph. Then every window node has* $\rho_W(v(t)) = T|\mathrm{rel}_S(v)|$*, and for every* T

$$s_\rho^W = \tfrac{1}{2} + \frac{|E_\uparrow|}{2|E_S|}, \quad E_\uparrow = \{(i,j) \in E_S : \mathrm{Src}(i) \neq \mathrm{Src}(j)\}.$$

Lean: `rhoBurn_eq`, `sBurn_eq`, `rhoZ_eq`, `sZ_eq`, `relSet_eq_iff_src`.

Sketch. After $p-1$ slices every finite distance is absorbed, so $w(s) \in \mathrm{rel}(v(t))$ iff $\mathrm{anc}_S(v) \cap \mathrm{anc}_S(w) \neq \emptyset$. For an edge (i, j), $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$ with equality iff $\mathrm{Src}(i) = \mathrm{Src}(j)$.

Table 1. Rel-sortability of 3-node summary graphs (with self-loops, $|E_S| = 5$). Finite: $\lim_T s_\rho(G_T)$ (Theorem 2); stationary: s_ρ^W (Theorem 5). Lean: `chain3_limit`, `chain3_stat`, `fork_limit`, `fork_stat`, `collider_limit`, `collider_stat`; 3-cycle by `reisach_theorem4`.

summary graph	finite G_T (limit)	stationary
chain $a \to b \to c$	0.7	0.5
fork $a \leftarrow b \to c$	0.7	0.5
collider $a \to c \leftarrow b$	0.7	0.7
3-cycle (6 edges)	0.5	0.5

Corollary 6 (stationary condition; graph classes)**.** $s_\rho^W = 1/2$ *iff every weakly connected component of* G_S *has exactly one source SCC. Strongly connected graphs, out-forests, chains and forks give* $1/2$*. In an in-forest (a river network),* $s_\rho^W = \frac{1}{2} + \#\{\text{edges into confluences}\}/(2|E_S|)$*, which exceeds* $1/2$ *iff there is a confluence.*

Lean: `sBurn_eq_half_iff`, `sZ_eq_half_iff`, `stat_half_of_outForest`, `statValue_inForest`, `statValue_inForest_gt_iff`, `collider_stat`, `fork_stat`.

Table 1 shows the two versions side by side. In the finite generator every summary DAG with the same m/p has the same value, so a benchmark built this way rewards relatives-sorting on chains and forks, where the stationary process gives no signal. For a discovery method applied to long stationary records, the stationary value is the relevant one, and it is what our real-data predictions use. We verified all statements of this section by exhaustive enumeration of all summary graphs with $p \leq 4$ and 10^4 random graphs with $p \leq 7$ (46,724 (graph, T) cases, 0 failures; Appendix B) before the Lean proofs were written.

4. Aggregation versus Subsampling

Let X be a stable linear SVAR with lags $\tau \geq 1$, independent innovations with diagonal covariance Σ, and summary graph G_S; $\Gamma(h)$ is its autocovariance and $A(1) = \sum_\tau A_\tau$. Consider block means $\bar{Y}_s = \frac{1}{D}\sum_{u<D} X(sD+u)$ and subsamples $Z_s = X(sD)$.

Proposition 7 (relatives survive any per-node filter)**.** *If* $\mathrm{anc}_S(i) \cap \mathrm{anc}_S(j) = \emptyset$ *then* X_i *and* X_j *are independent at all lags, and so are any per-node, time-invariant (possibly nonlinear) transforms of them, including* $\bar{Y}$ *and* Z *for every* D*. If the ancestor sets intersect,* $\Gamma(0)_{ij}$*,* $\mathrm{Cov}(\bar{Y})_{ij}$ *and the long-run covariance are nonzero for Lebesgue-almost every parameter.*

Lean: not machine-checked (needs the SVAR moving-average representation and real-analytic genericity).

So the population relatives of contemporaneous values are rel_S for every observation scheme, and the population relatives sort equals the stationary value of Theorem 5 (on

non-self-loop edges), not the finite-G_T value. Var- and R^2-sortability behave differently.

Theorem 8 (block means vs subsampling)**.** *(a)* $\|D\,\mathrm{Cov}(\bar{Y}) - S_{LR}\| \le \frac{1}{D}\sum_h |h|\|\Gamma(h)\|$, *with* $S_{LR} = (I - A(1))^{-1}\Sigma(I - A(1))^{-\top}$. *(b) If* G_{S} *is acyclic apart from self-loops,* S_{LR} *is the covariance of a static SEM on* G_{S} *with weights* $A(1)_{ji}/(1-\lambda_j)$ *and noise variances* $\sigma_j^2/(1-\lambda_j)^2$, $\lambda_j = A(1)_{jj} < 1$. *(c)* $\mathrm{Cov}(Z_s) = \Gamma(0)$ *for every* D. *(d) Hence, for large* D, *block-mean* R^2*-sortability is* 0 *on out-stars and* 1 *on in-stars with* ≥ 2 *arms, for all weights; under subsampling every contemporaneous criterion keeps its* $\Gamma(0)$ *value.*

Lean: (a) counting identity `fejer_identity`*; (b)* `longrun_factor`, `selfloop_sum_lt_one`*; the probabilistic steps, the rate and (c)–(d) are not machine-checked.*

Block means and subsampling therefore converge to different contemporaneous objects (a long-run SEM versus the stationary snapshot), and block means can reverse the R^2 order a snapshot has: in our exact computations on out-stars, subsampled data are R^2-sorted correctly on average (0.84) while block means drift to 0 (Appendix C). Rel-sortability is the only one of the three criteria that depends on the graph alone and is invariant to the observation scheme, which is why we study it further.

5. Finite-Sample Theory

We now ask whether the population relatives can be estimated. X is a zero-mean stationary Gaussian process with summable autocovariances $\gamma_{ab}(h)$, standardised so that $\gamma_{aa}(0) = 1$, observed at n times. Two nodes are declared relatives if the Fisher-z test of the sample correlation $\hat{r}_{ab}$ rejects at level α with Bonferroni over $m = d(d-1)/2$ pairs, i.e. $|\hat{r}_{ab}| > \tanh(z_c/\sqrt{n'-3})$, $z_c = \Phi^{-1}(1-\alpha/2m)$; $\hat{\rho}(v) = 1 + \#\text{declared partners}$. The *nominal* test uses $n' = n$. Let $B_{ab} = \sum_h |\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)|$ (for an independent pair, the Bartlett sum $\sum_h \rho_a(h)\rho_b(h)$).

F1: recovery and effective sample size.

Theorem 9 (recovery bound)**.** *For* $x, y > 0$ *let* $\varepsilon_{ab} = \sqrt{2B_{ab}x/n} + \kappa_2 x/n + 2\kappa_1 y/n$, $\varepsilon_d = \sqrt{2D_\gamma y/n} + 4\kappa_1 y/n < 1$ *and* $\delta_{ab} = (\varepsilon_{ab} + |\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$, *with autocovariance constants* $D_\gamma, \kappa_1, \kappa_2$ *defined in Appendix D. If thresholds separate* ($\delta_{ab} \le \tau_{ab}$ *on non-relatives and* $|\rho_{ab}| - \delta_{ab} > \tau_{ab}$ *on relatives), the declared set equals the true one, and hence* $\hat{s} = s_\rho^{\mathrm{pop}}$, *with probability at least* $1 - d(d-1)e^{-x} - 4de^{-y}$. *For a common threshold* $\tau < \rho_{\min}$ *the failure probability decays as* $\exp(-n\min(\tau, \rho_{\min} - \tau)^2/(2(\sqrt{B} + \rho_{\max}\sqrt{D_\gamma})^2))$, *so* n/B *plays the role of an effective sample size.*

Lean: deterministic core `thm15_det`, `decide_eq_target`, `corr_perturb`, `sOf_cntDecl_congr`*; Chernoff and trace algebra* `chernoff_choice`, `phi_le`, `lemma12_trace`, `lemma12_le_B`*. The Gaussian tail bound itself is not machine-checked.*

For i.i.d. snapshots $B = 1 + \rho^2$; under strong persistence B can be in the tens, and so can the loss in effective sample size. The proof combines a Gaussian-chaos tail bound (the Gaussian case of Hsu et al., 2012, Prop. 1; see also Laurent & Massart, 2000), an exact finite-n variance identity for $\hat{r}$, and a deterministic perturbation lemma.

Proposition 10 (the nominal test is inconsistent; asymptotic)**.** *Assuming the standard CLT for quadratic forms of stationary Gaussian processes, a non-relative pair is declared by the nominal test with probability* $\to 2\bar{\Phi}(z_c/\sqrt{B_{ab}})$, *which does not vanish. With* $n_{\mathrm{eff}} = n(1-\rho^2)^2/W_{ab}$ (W *the delta-method Bartlett variance) the false-positive rate tends to* α/m.

Lean: not machine-checked (relies on a CLT not available in Mathlib).

F2: saturation by a latent driver. Let $X_v = Y_v + b_v C$ with C a stationary unit-variance process independent of the VAR Y, and $\ell_v = b_v^2/\mathrm{Var}X_v$ the driver's share of variance.

Proposition 11 (saturation)**.** *For summary non-relatives,* $\rho_{uv} = \mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v} =: \kappa_{uv}$. *If all* $b_v \ne 0$, *every pair is a population relative and the population rel-sortability is exactly* $1/2$; $P(\hat{s}_n \ne \frac{1}{2}) \le (d^2+3d)e^{-cn}$ *with* $c \approx \kappa_{\min}^2/(18(\sqrt{B} + \sqrt{D_\gamma})^2)$. *Asymptotically, a pair is detected with probability* $> 1/2$ *once* $n > n_{1/2} = 3 + (z_c/\operatorname{atanh}\kappa)^2 \approx z_c^2/\ell^2$ *(nominal test; for the effective test, the same bound on* n_{eff}*).*

Lean: `prop21_corr_std`, `sOf_saturated`, `abs_sOf_sub_half_le`*, location of* $n_{1/2}$*:* `ncrit_eq_iff`*; the probability statements are not machine-checked.*

$n_{1/2}$ does not depend on autocorrelation, which only sets the width of the transition. A driver with share $\ell = 0.1$ saturates at $n \approx 10^3$, and one with $\ell = 0.64$ at $n \approx 25$. Since $\tau_n \to 0$ for any consistent test, a relatives count on a long record can recover the summary relatives only inside a window $\max_{\text{non-rel}}|\kappa| < \tau_n < \min_{\text{rel}}|\rho|$, which closes at n of order $n_{1/2}$. Autocorrelation (Proposition 10) and a latent driver are two separate failure modes; both push $\hat{s}$ towards $1/2$.

F3: conditioning on a proxy. Let $Z = \sqrt{r}\,C + \sqrt{1-r}\,e$ with reliability $r = \mathrm{corr}(Z, C)^2$, and let π_{uv} be the partial correlation given Z.

Proposition 12 (conditioning)**.** *With* ρ^Y *the driver-free correlation,*

$$\pi_{uv} = \frac{\rho^Y_{uv}\sqrt{(1-\ell_u)(1-\ell_v)} + (1-r)\,\mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v}}{\sqrt{(1-r\ell_u)(1-r\ell_v)}}.$$

At $r = 1$ the conditional relatives are the summary relatives and Theorem 5 is restored. For non-relatives with $\ell_u, \ell_v \in (0, 1)$, $|\pi_{uv}|$ is strictly decreasing in r and vanishes only at $r = 1$. Spurious conditional relatives therefore appear for every $r < 1$ once n is large; fewer than half of the non-relative pairs are declared only if $r > r_{\text{crit}} = 1 - \frac{\tau'(1-\ell)}{\ell(1-\tau')}$, $\tau' = \tanh(z_c/\sqrt{n'-4})$ (equal shares). If the driver reaches nodes through different filters (e.g. travel times k_v), a scalar proxy leaves a nonzero residual even at $r = 1$. The reliability is identified by $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$.

Lean: `prop31_std`, `cor32`, `absPi_strictAntiOn`, `absPi_eq_zero_iff`, `rcrit_iff`, `thm35_bracket`, `pc_lipschitz`, `prop36_ar1_ne_zero`, `prop36_rank_one`, `prop38_ratio`.

Formalisation found four missing nondegeneracy hypotheses in our own earlier statements (e.g. $b_u b_v \neq 0$ for strict monotonicity); they are included above and listed in Appendix F. For $\ell = 0.64$ and a typical threshold $\tau' = 0.082$, $r_{\text{crit}} = 0.95$.

The non-asymptotic bounds are valid but loose (factors 3–13 in n in simulation, Appendix E); the asymptotic statements matched every simulation to within 0.03 and are what we use for prediction below.

6. Machine-Checked Proofs

All results of Section 3 and the deterministic and algebraic parts of Sections 4 and 5 are formalised in Lean 4 (de Moura & Ullrich, 2021) with Mathlib (The mathlib Community, 2020) (3,343 lines). The axiom report lists 116 theorems, each depending only on `propext`, `Classical.choice` and `Quot.sound`; the sources contain no `sorry`, `admit` or `axiom`. Relatives, s_ρ, G_T, burn-in windows and the infinite-past graph are defined from scratch following Reisach et al. (2026); the one modelling convention is that "a directed path exists" is read as reflexive-transitive reachability. Not formalised: Gaussian tail bounds, Isserlis' theorem, the CLT, the SVAR moving-average representation and genericity (Proposition 7). These are the parts a referee must check by hand; Appendix F maps every statement to its Lean name and status.

7. Pre-Registered Real-Data Evidence

Protocol. We ran four pre-registered studies (P1–P9, then E1/E2). Each prediction, threshold, estimator, seed and kill rule was committed to version control before any dependence statistic of the data was computed (timeline in Appendix H). The predicted value for an observed node set S is the stationary value (Theorem 5) computed in the full known graph with unobserved nodes latent, averaged over the induced non-self-loop edges ($\tilde{s}$). The estimator is the relatives count of Section 5 at $\alpha = 0.01$ with Bonferroni; orientation accuracy is the mean edge score of $\hat{\rho}$. CIs are cluster bootstraps (2,000 resamples).

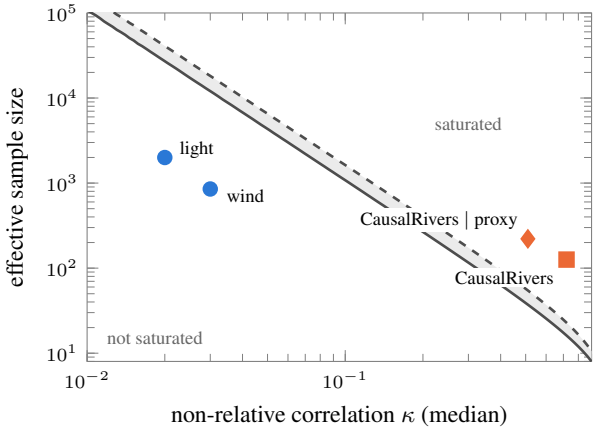


Figure 2. Saturation regime of Proposition 11. Curves: onset $n_{1/2}$ for $z_c = 3.29$ ($d = 5$, solid) and 4.04 ($d = 20$, dashed); above them a non-relative pair is detected with probability $> 1/2$. Points: medians from the real-data studies (CausalRivers unconditioned and given the regional proxy, κ replaced by $|\pi|$; Causal Chambers wind and light tunnel). Values are in Section 7 and Appendix G.

CausalRivers: a negative result the theory explains. River networks are in-forests, so Corollary 6 predicts $\tilde{s} > 1/2$ for subgraphs with confluences (mean 0.82). The estimated relatives sort carried no orientation information (P2, P3 fail; Table 2) because 98% of non-relative station pairs tested dependent. Read through F2 (post hoc for this study): non-relative correlations $\kappa \approx 0.64$–0.74 give $n_{1/2} \approx 20$–27, while the records have $n = 377$–$3{,}652$ steps; the study sat two orders of magnitude inside the saturated regime (Figure 2), where Proposition 11 predicts $\hat{s} = 1/2$. Regional precipitation acts as a latent driver; adding it to the summary graph leaves one source per component, and Corollary 6 then predicts exactly $1/2$. The second study pre-registered the F3 remedy (condition on a leave-out regional mean). It failed (P4: 0.74), and F3 accounts for it: the triad estimate of reliability is $\hat{r} \approx 0.62$, below the $r_{\text{crit}} \approx 0.87$ required at the observed n_{eff} (only 26% of pairs exceed it), and the plug-in prediction of the post-conditioning dependent fraction is 0.743 (0.736) against an observed 0.741 (0.735) at 12 h (24 h). This check is in-sample, because $\hat{r}$ and $\hat{\ell}$ come from the tested correlations. Post hoc, conditioning on 10 principal components still leaves 49% of non-relative pairs dependent, consistent with a driver of rank > 1 (Proposition 12).

Causal Chambers: designed independent sources. The third study chose data in which Corollary 6's non-trivial case holds by design: in the wind-tunnel random-walk runs (Gamella et al., 2025) five actuators are independent experimenter-generated random walks, so they are separate source SCCs; 25 of 26 edges are predicted to be strictly

Table 2. Pre-registered tests. Pass/fail is by the pre-registered point rule. Brackets: 95% cluster-bootstrap CI. "non-rel. dep.": fraction of graph non-relative pairs declared dependent. "Acc.": orientation accuracy on edges predicted to be strictly sorted.

study (units)	test (threshold)	result (12 h / 24 h)	verdict
CausalRivers, marginal (120 subgraphs)	P1: confluence subgraphs have $\tilde{s} > 1/2$	112 of 114	fails as worded[a]
	P2: Acc. > 0.5, CI excl. 0.5	$0.501\,[0.492, 0.509]$ / $0.498\,[0.489, 0.507]$	**fail**
	P3: Spearman$(\tilde{s}, \text{acc.}) \geq 0.3$	-0.018 / 0.018	**fail**
	(non-rel. dep., nominal test)	0.984 / 0.978	kill rule triggered
CausalRivers, conditioned on regional proxy (120 subgraphs)	P4: non-rel. dep. < 0.5 given proxy	0.741 / 0.735	**fail**
	P5: Acc. > 0.5, CI excl. 0.5	$0.539\,[0.513, 0.565]$ / $0.540\,[0.514, 0.567]$	pass, not interp.[b]
	P6: Spearman ≥ 0.3	-0.113 / -0.121	**fail**
Causal Chambers, wind tunnel (15 runs, primary)	P7: non-rel. dep. < 0.5	$0.039\,[0.031, 0.047]$	pass
	P8: Acc. > 0.5, CI excl. 0.5	$0.656\,[0.637, 0.675]$	pass
	P9: Spearman ≥ 0.3 (300 subgraphs)	$0.267\,[0.163, 0.368]$	**fail**
light tunnel (10 segments, replication)	R7 / R8 / R9 (same thresholds)	0.000 / 1.000 / $0.779\,[0.696, 0.864]$	pass (replication)[c]
E1, light tunnel (10 units, G_{lin})	E1a: non-rel. dep. < 0.5; det. ≥ 0.8	0.000; $0.969\,[0.945, 0.992]$	pass
	E1b: $\Delta \leq -0.10$, Holm $p \leq 0.05$, all 8	$\Delta = -0.502$ to -0.533 (10/10)	pass (8/8)[d]
	E1c: as E1b, detected edges	$\Delta = -0.500$	pass (8/8)
E2, wind regimes (64 units; D=1, lags L=30 vs. 0)	Q7: non-rel. dep. < 0.5	$0.003\,[0.001, 0.005]$	pass
	Q8: Acc. > 0.5, CI excl. 0.5	$0.553\,[0.527, 0.579]$	pass
	Q9: Spearman ≥ 0.3 (1,280 subgr.)	$0.243\,[0.193, 0.294]$	**fail**
	Q10: detection gain, CI lower bound > 0	$+0.069\,[0.060, 0.079]$	pass
	E2 success (Q7–Q10 all)		**not met**

[a]The two exceptions are split-and-rejoin channels with a single source, for which the theorem itself predicts $1/2$. [b]Kill rule (P4 failed at both resolutions): P5 is reported, not interpreted. [c]Pre-registered as not counting towards a positive demonstration (i.i.d. inputs, weak time-series aspect). [d]G_{lin} is the pre-registered primary graph. It omits six real causal edges that, under a symmetry lemma and an assumption of ideal polariser sensors, carry no linear dependence. On the published graph the theorem still beats all eight overall, but not on its strictly ordered edges (Section 7, Appendix I).

Table 3. Out-of-sample F-theory check on the Causal Chambers: predictions from the first half of each unit, observed rates on the second half. Accuracy is predicted by Monte Carlo of independent pair decisions.

	wind tunnel		light tunnel	
	pred.	obs.	pred.	obs.
non-relatives declared (F2)	0.031	0.028	0.006	0.000
relatives detected (F1)	0.360	0.343	0.679	0.681
orientation accuracy	0.630	0.620	0.999	1.000

sorted. Saturation is absent (P7, 0.039) and the relatives count orients confluence edges above chance (P8, 0.656). The graded agreement P9 missed its threshold ($0.267 < 0.3$; CI contains 0.3), so by our own pre-registered definition the study is *not* a positive real-data demonstration. The reason is F1, not a driver: 16 of the 26 ground-truth edges are essentially undetectable at lag 0 at $n \approx 10^3$ (smallest relative correlation ≈ 0.002), which compresses the estimated counts. The light-tunnel replication with i.i.d. inputs passes all three criteria (Table 2). F-theory predictions computed from the first half of each unit (plug-in correlations and autocorrelations, Fisher-z asymptotics of Propositions 10 and 11) and evaluated on the second half are accurate to within 0.02 (Table 3). In particular F2 predicts no saturation: $\hat{\kappa} \approx 0.03$ (wind) and 0.02 (light) give $n_{1/2} \approx 15{,}600$ and 33,800, far above the available $n_{\text{eff}} \approx 430$–$1{,}000$.

A measured driver (ambient pressure, shared by four barometers with no graph edge) is detected exactly where F2 predicts ($\kappa = 0.19$–0.32, $n_{1/2} = 145$–$426 < n_{\text{eff}}$), and conditioning on it removes it (dependent fraction $0.039 \to 0.001$). Two caveats: the wind tunnel graph is bipartite, so the prediction coincides with "sensors have more relatives than actuators" (the motivation for E1, below); and all units of a chamber share one device.

E1: does the theorem beat naive predictors? Pre-registered (Appendix H) to answer the bipartite caveat. The test unit is an edge instance: the theorem's value $s_\rho \in \{0, \frac{1}{2}, 1\}$ against the observed value from the estimated counts, with absolute loss, on the instances where the theorem and a competitor disagree. The eight competitors are the sensor indicator; ancestor count, in-degree and depth (each on the full graph and on the induced subgraph); and the finite-T limit "every edge is 1". A competitor counts as beaten if $\Delta \leq -0.10$ with an exact sign-flip test over units and Holm correction over the eight. The primary data are the light-tunnel series `citris_1` (10 units), the only real time series we found whose ground truth has several layers among measured variables (red→green→blue→sensors). **It passes 8/8.** The pre-registered power prerequisites hold (non-relatives declared 0.000, relatives detected 0.969); $\Delta = -0.502$ (sensor indicator) to -0.533, in 10/10 units

each, Holm $p = 0.0078$, the smallest attainable; restricted to pairs declared dependent, $\Delta = -0.500$. The theorem's value equals the observed one on 99.2% of 1,662 instances, the competitors' on 11–20%; its three anti-sorted predictions (red and blue $\to \mathrm{pol}_2$, true edges whose head has fewer relatives) hold in 99.1% of 110 instances. The same test on 45 i.i.d. structural-model units (pooled, 8,734 instances) also passes 8/8 ($\Delta \approx -0.49$, 45/45 units); the chain gives $1/2$ everywhere and the collider 1 into the collider, as Corollary 6 predicts. Four statements bound this result. *(i) The primary prediction uses a refined graph.* It removes six real causal edges (red, blue $\to \mathrm{pol}_2$; the polarisers $\to$ two sensors) because, by a symmetry lemma (Appendix I) that assumes ideal polariser sensors, they carry no linear dependence at any lag. The claim is therefore only that the theory predicts the *linearly dependent* relatives. On the published graph the theorem still beats all eight overall ($\Delta = -0.32$ to -0.42), but on the 442 edges where it predicts strict order the sensor indicator is significantly better ($\Delta = +0.453$, reverse Holm $p = 0.002$), and the theorem's exact-match rate (71.4%) is below that of a constant-$1/2$ rule (81.5%, post hoc). *(ii) Most of the separation comes from ties.* Single-source components are predicted $1/2$, which none of the naive rules can express: 1,365 of the 1,662 predictions are $1/2$. Strict $0/1$ predictions are fewer (297 on `citris_1`, 2,768 on the snapshot sets). On `citris_1` the theorem also wins on them against all eight ($\Delta = -0.54$ to -0.99), where an estimator without power would favour the competitors. Post hoc, a constant-$1/2$ rule matches 81.5% of `citris_1` (68.8% of the snapshot sets) against 99.2% (92.3%). *(iii) The evidence is one device in a favourable regime*: a light tunnel with strongly responding sensors and detection of 0.97; the snapshot sets are also light-tunnel data. *(iv) The wind tunnel does not replicate the strength.* On the 64 held-out regimes the theorem passes the same test (primary $\Delta = -0.225$, secondary -0.203), but detection is only 0.33–0.59 (limited by F1), so part of this is power-induced. On the primary configuration a constant-$1/2$ rule has lower absolute error than the theorem (0.311 vs. 0.363); at $D{=}10$ the two are close (0.313 vs. 0.294).

E2: lagged relatives. E2 tests whether a relatives test that uses cross-correlations at lags $|h| \le L$ (Bonferroni over lags; same population relatives under lag-L faithfulness, Appendix J) recovers the weak lag-0 edges behind P9, on 64 held-out regimes (primary: first differences, $L = 30$, against $L = 0$). Success required Q7–Q10 to pass. **It does not**, because Q9 fails (Table 2): false positives stay rare (Q7 0.003), the accuracy on edges predicted 1 is 0.553 (Q8), detection rises from 0.328 to 0.398 ($+0.069$, Q10), but the graded agreement is $0.243 < 0.3$ (Q9). The secondary configuration (block means, $L = 3$) fails Q9 (0.202) and Q10 ($-0.007\,[-0.012, -0.003]$): the Bonferroni cost exceeds the gain. On `citris_1` lags add $+0.008\,[0, 0.024]$ (detection is already 0.97). The lag-0 estimator of P7–P9, rerun on these held-out regimes, again fails the graded criterion (0.202 at $D{=}10$, 0.120 at $D{=}1$): the P9 failure repeats, and its original verdict (0.267) stands. E2 does not rescue P9. Re-drawing only the subgraph sample on the P9 data moved that statistic from 0.267 to 0.338 (development, not a verdict); we report this as variability of the statistic of about 0.07, not as a pass. Lags recover some edges (load_in $\to$ upwind pressure $0.56 \to 0.98$) and lose others to the larger threshold ($\mathrm{pot}_1 \to \mathrm{signal}_1$ $0.36 \to 0.16$); every microphone edge stays undetected, so the F1 shortfall is not mainly a lag problem. Out-of-sample F1/F2 checks pass in all six configurations (Appendix J).

8. Discussion and Limitations

For benchmark design. The finite time-unrolled generator is rel-sortable above $1/2$ for every summary graph that is not a union of strongly connected components (Corollary 3), and part of the excess comes only from the start at $t = 1$. Benchmarks should either use strongly connected summary graphs or a burn-in of at least $p - 1$ slices, after which the value is the graph property of Theorem 5, for every T. Reporting $|E_\times|/|E_\mathrm{S}|$ or $|E_\uparrow|/|E_\mathrm{S}|$ alongside a benchmark costs nothing.

For real data. Population rel-sortability above $1/2$ does not imply that a relatives count can be estimated. Two conditions are needed, and F1–F3 quantify both: the absence of a latent common driver with share ℓ such that $n \gtrsim z_c^2/\ell^2$ (or a near-perfect, rank-one proxy for it), and relatives that are detectable at lag 0 at the available n_eff. CausalRivers violates the first, the wind tunnel partly the second. A cheap diagnostic follows: before relying on relatives-based orderings (or before reading a benchmark result as evidence that a method uses them), estimate the typical correlation $\hat\kappa$ of pairs believed to be unrelated and compare n_eff with $3 + (z_c/\operatorname{atanh}\hat\kappa)^2$; if n_eff is far above it, the relatives count will be saturated whatever the graph. A practical consequence for method developers is that a time-series discovery method cannot gain much from rel-sortability on real records with a shared driver, even when the summary graph would make it informative in population.

Open questions. Summary graphs without self-loops, contemporaneous effects at the native time step, and nonlinear mechanisms (where Proposition 7 still gives the zero pattern but faithfulness can fail, as the Malus-law pairs of the light tunnel show) are not covered. Lagged cross-correlations raise detection at fine resolution (E2) but did not repair the graded agreement of P9; weak, nonlinear or undetectable effects remain the limiting factor on the wind tunnel.

Limitations. The graph results assume self-loops at every node (without them a single edge gives $1/2$ instead of $2/3$;

Appendix A). Section 4 covers linear SVARs only. The finite-sample bounds are loose, and the asymptotic statements rest on a CLT that we cite rather than prove. Proposition 7 and the probabilistic parts of F1–F3 are proved on paper only. The real-data evidence comes from two benchmark families, and the positive part from one physical device per chamber (E1 uses light tunnels only). The strongest result (E1) holds for linearly dependent relatives, relies on an assumption about ideal polariser sensors, and is carried mainly by predicted ties; faithfulness to linear dependence fails by design for six real edges. The graded criteria P2, P3, P4, P6, P9 and Q9 failed.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Artefact Statement

Lean code, verification scripts, analysis scripts, preregistration documents and run logs are provided as supplementary material (anonymised). Raw data are public (CausalRivers; Causal Chambers, CC BY 4.0) and are not redistributed.

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Appendix overview

Appendix A gives full proofs of Section 3, Appendix B the exhaustive enumeration check, Appendix C the proofs of Section 4, Appendix D the proofs of Section 5, Appendix E the simulation checks of F1–F3, Appendix F the map from statements to Lean names, Appendix G details of the real-data studies, Appendix H the pre-registration timeline and deviations, and Appendices I and J the experiments E1 and E2 in full.

Status labels. *Machine-checked*: a Lean theorem with no `sorry` and only the standard axioms. *Proved*: a complete paper proof given here. *Asymptotic*: the argument uses the standard central limit theorem for quadratic forms of stationary Gaussian processes with summable covariances, which we cite and do not prove.

A. Proofs for Section 3

Throughout, $G_S = (V, E_S)$ has a self-loop at every node, $p = |V| \geq 2$, and $d(u, v)$ is as in Section 2. We use two facts of Reisach et al. (2026), proved there for DAGs: (Thm. 1) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subseteq \mathrm{rel}(y)$; (Lemma 1, eq. (10)) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subsetneq \mathrm{rel}(y)$ iff $\mathrm{roots}(y) \setminus \mathrm{roots}(x) \neq \emptyset$, where $\mathrm{roots}(y) = \{x \in \mathrm{anc}(y) : \mathrm{anc}(x) = \{x\}\}$. Both are also machine-checked independently (`relSet_subset_of_edge`, and the direct characterisation `isRel_unrolled_iff`).

Lemma 13 (ancestors in G_T)**.** *$u(s) \in \mathrm{anc}(v(t))$ iff $s \leq t$ and $d(u, v) \leq t - s$.*

Lean: `unrolled_reach_iff`, `unrolled_reach_iff_dist`.

Proof. Every edge of G_T increases time by one, so a directed path from $u(s)$ to $v(t)$ has $t - s$ edges, and its projection to V is a walk of length $t - s$ from u to v in G_S (self-loops allowed as steps). Such a walk exists iff $d(u, v) \leq t - s$: a walk contains a path, so its length is at least $d(u, v)$; conversely, pad a shortest path with $t - s - d(u, v)$ self-loop steps at u. Every such walk lifts to a directed path in G_T. □

Lemma 14 (roots)**.** $\mathrm{roots}(v(t)) = \{u(1) : d(u, v) \leq t - 1\}$, *and* $\mathrm{desc}(u(1)) = \{w(s) : d(u, w) \leq s - 1\}$.

Lean: `roots_unrolled_iff`, `isRoot_unrolled_iff`, `desc_root_unrolled_iff`.

Proof. A node $u(s)$ with $s \geq 2$ has the parent $u(s - 1)$ through the self-loop, so it is not a root; $u(1)$ has no parent. Apply Lemma 13 with $s = 1$, in both directions. □

Proof of Theorem 1. $i(t)$ is a parent of $j(t{+}1)$, so $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$ and $s_\rho(e) \in \{1/2, 1\}$, with value 1 iff the inclusion is strict, iff $\mathrm{roots}(j(t{+}1)) \setminus \mathrm{roots}(i(t)) \neq \emptyset$. By Lemma 14 this set is $\{u(1) : d(u, j) \leq t,\ d(u, i) > t - 1\}$, and since d is integer-valued or ∞, $d(u, i) > t - 1$ iff $d(u, i) \geq t$. □

Proof of Proposition 4 (and the general count). Since every ancestor of y descends from a root of y that is itself an ancestor, $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{roots}(y))$. By Lemma 14, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u, v) \leq t - 1$ and $d(u, w) \leq s - 1$, i.e. iff $s \geq 1 + m_t(v, w)$ with $m_t(v, w) = \min\{d(u, w) : d(u, v) \leq t - 1\}$. Counting $s \in \{1, \ldots, T\}$ gives $\rho(v(t)) = \sum_w \max(T - m_t(v, w), 0)$ (`rho_unrolled_eq`). For $t \geq p$ every finite $d(u, v)$ is at most $p - 1 \leq t - 1$, so the admissible u are $\mathrm{anc}_S(v)$ and $m_t(v, w) = d(\mathrm{anc}_S(v), w)$, finite iff $w \in \mathrm{rel}_S(v)$ and then at most $p - 1 < t \leq T$. (The Lean statement `rho_unrolled_large_t` needs only $t \geq p$; an earlier hypothesis $T \geq p$ is redundant.) □

Proof of Theorem 2. Crossing edge $(i, j) \in E_\times$: j does not reach i, so $d(j, i) = \infty$, and $u = j$ gives $d(j, j) = 0 \leq t \leq \infty$ for every t: all $T - 1$ copies score 1. Edge in $E_\circ$: i and j reach each other (or $i = j$), so for every u, $d(u, i) < \infty$ iff $d(u, j) < \infty$. If the condition of Theorem 1 held with $d(u, i) = \infty$ then $d(u, j) = \infty > t$, a contradiction; hence it forces $t \leq d(u, i) \leq p - 1$, and at most $N_T(i, j) \leq \min(p - 1, T - 1)$ copies score 1. Summing over the $(T - 1)|E_S|$ edges,

$$s_\rho(G_T) = \tfrac{1}{2} + \frac{1}{2(T-1)|E_S|}\Big[(T-1)|E_\times| + \sum_{(i,j)\in E_\circ} N_T(i, j)\Big],$$

which gives both bounds; let $T \to \infty$. □

Proof of Corollary 3. $E_\times = \emptyset$ iff no edge joins two SCCs. A weakly connected component made of two or more SCCs is connected by undirected edges, one of which joins two different SCCs; conversely a strongly connected component has no

crossing edge. The uniform bound is Theorem 2 with $|E_\times| \geq 1$. For the DAG case, every SCC is a singleton, $E_\times$ consists of the m non-loop edges, and $|E_{\mathrm{S}}| = m + p$. □

Proof of Theorem 5. In G_{B+T}, by Lemma 14 shifted by B, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u,v) \leq t-1$ and $d(u,w) \leq s-1$. For window nodes $t-1, s-1 \geq B \geq p-1$, and every finite distance is at most $p-1$, so the condition is $u \in \mathrm{anc}_S(v) \cap \mathrm{anc}_S(w)$, independent of s,t. Hence $\mathrm{rel}_W(v(t)) = \{w(s) \in W : w \in \mathrm{rel}_S(v)\}$, of size $T|\mathrm{rel}_S(v)|$. On $V \times \mathbb{Z}$, Lemma 13 holds verbatim and the common ancestor can be taken arbitrarily early, giving the same set. Each copy of (i,j) therefore compares $|\mathrm{rel}_S(i)|$ with $|\mathrm{rel}_S(j)|$; as $i \in \mathrm{anc}_S(j)$, $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$, so the score is 1 iff $\mathrm{rel}_S(i) \neq \mathrm{rel}_S(j)$, and $1/2$ for self-loops. Source SCCs: $\mathrm{rel}_S(j) \setminus \mathrm{rel}_S(i) \neq \emptyset$ iff $\mathrm{anc}_S(j) \not\subseteq \mathrm{rel}_S(i)$ (if $z \in \mathrm{rel}_S(i)$ then $\mathrm{desc}_S(z) \subseteq \mathrm{rel}_S(i)$). Every node of $\mathrm{anc}_S(j)$ is reached from some $\sigma \in \mathrm{Src}(j)$, and $\mathrm{desc}_S(\sigma) \subseteq \mathrm{rel}_S(i)$ whenever $\sigma \subseteq \mathrm{rel}_S(i)$. A source SCC meets $\mathrm{rel}_S(i)$ iff it reaches i (its only ancestors are its own nodes), and then it is contained in $\mathrm{rel}_S(i)$. So the condition is $\mathrm{Src}(j) \not\subseteq \mathrm{Src}(i)$, which, as $\mathrm{Src}(i) \subseteq \mathrm{Src}(j)$, is $\mathrm{Src}(i) \neq \mathrm{Src}(j)$. □

Proof of Corollary 6. If each weak component C has a unique source SCC σ_C, then $\mathrm{Src}(v) = \{\sigma_C\}$ on C and $E_\uparrow = \emptyset$. Conversely, if $E_\uparrow = \emptyset$, Src is constant along edges and hence on each weak component; a node v of a source SCC σ has $\mathrm{Src}(v) = \{\sigma\}$, so there is only one. Out-forests (at most one parent besides itself; acyclicity is not needed, `stat_half_of_outForest`) and strongly connected graphs (`stat_half_of_strong`) have one source SCC per component. In an in-forest (acyclic apart from self-loops, out-degree at most one), if $u \in \mathrm{anc}(v)$ then $\mathrm{desc}(u)$ is the unique downstream path through v, so $\mathrm{rel}(v) = \mathrm{anc}(v) \cup \mathrm{desc}(v)$ (`inForest_isRel_iff`); the source SCCs are the headwaters, and $\mathrm{Src}(j) = \mathrm{Src}(i) \cup \bigcup_k \mathrm{Src}(k)$ over the other parents k of j, with disjoint headwater sets on different branches, so $\mathrm{Src}(j) \neq \mathrm{Src}(i)$ iff j is a confluence (`inForest_up_iff`). Out-degree at most one alone is not enough: for $a \rightleftarrows b$ with $c \to a$, a has two parents but the only source SCC is $\{c\}$. □

Remark 15 (self-loops). Without self-loops the lemmas fail: for the single edge $0 \to 1$ without loops, $s_\rho(G_T) = 1/2$ for every $T \geq 2$ (`edgeNoLoops_sRho`); with loops the limit is $2/3$ (`edgeLoops_limit`).

B. Exhaustive Enumeration Check

Before formalisation, every statement of Section 3 was checked by brute force (bitset relatives with the definitions of Reisach et al., 2026) on all $4 + 64 + 4096$ summary graphs with $p = 2, 3, 4$ (all $T \leq 9$) and on 10,000 random graphs ($p = 2$–7; Erdős–Rényi, DAG, in-forest, out-forest, chain with back edge and block-SCC families; three values of $T \leq 14$ each, fixed seed). In total 46,724 (graph, T) cases and 1,895,609 edges were checked with zero failures of Theorem 1, Proposition 4, the bounds of Theorem 2, Theorem 5 (with $B = p-1$ and $B = 2p+1$) and the source-SCC characterisation. The conjecture we had pre-registered ("an edge is strictly sorted in the limit iff $|\mathrm{rel}_S(j)| > |\mathrm{rel}_S(i)|$") disagreed with the exact G_T limit in 9,603 random graphs and never with the stationary value; the conjecture is Theorem 5, not the limit of G_T.

C. Proofs for Section 4

Assumptions: $X(t) = \sum_{\tau=1}^{L} A_\tau X(t-\tau) + \varepsilon(t)$, $t \in \mathbb{Z}$, i.i.d. innovations with diagonal positive-definite covariance Σ, stable (all roots of $\det(I - \sum_\tau A_\tau z^\tau)$ outside the unit disc), stationary solution; the summary graph has $i \to j$ iff $(A_\tau)_{ji} \neq 0$ for some τ.

Proof of Proposition 7. By the MA(∞) representation $X(t) = \sum_k \Psi_k \varepsilon(t-k)$, and because Ψ_k is supported on the reachability relation, $X_i(t)$ is a measurable function of $\{\varepsilon_k(s) : k \in \mathrm{anc}_S(i)\}$. Disjoint ancestor sets mean independent innovation sets, so the processes X_i and X_j are independent, and so are any per-node transforms. For the generic part, each covariance entry is real-analytic in (A, Σ) on the connected open set of stable parameters with the given support, hence either identically zero or nonzero almost everywhere. With all coefficients positive, including the self-loop at a common ancestor k, all Ψ entries are nonnegative and $\Psi_{ik}(m), \Psi_{jk}(m) > 0$ for all large m (self-loops pad the walks), so $\Gamma(0)_{ij} > 0$; the same holds for $\Gamma(h)$, $h \geq 0$, and therefore for the Fejér sums and S_{LR}. "Generic" does not mean "large": with little autocorrelation, $\Gamma(0)_{ij}$ along a lagged edge can be tiny (down to 10^{-14} in our exact computations; exactly 0 without autocorrelation). □

Proof of Theorem 8. (a) $D^2 \mathrm{Cov}(\bar{Y}) = \sum_{u,w<D} \Gamma(u-w) = D\Gamma(0) + \sum_{h=1}^{D-1} (D-h)(\Gamma(h) + \Gamma(-h))$ (the counting identity is `fejer_identity`). The stable VAR has $\|\Gamma(h)\| \leq C r^{|h|}$, $r < 1$, so $\sum_h |h| \|\Gamma(h)\| < \infty$ and $S_{LR} =$

$\sum_h \Gamma(h) = \Psi(1)\Sigma\Psi(1)^\top$ with $\Psi(1) = (I - A(1))^{-1}$. Then $S_{LR} - D\,\mathrm{Cov}(\bar{Y}) = \sum_{|h|\geq D}\Gamma(h) + \sum_{|h|<D}\frac{|h|}{D}\Gamma(h)$, and $1 \leq |h|/D$ in the first sum. (b) Write $I - A(1) = (I - \Lambda) - N$ with Λ the diagonal and N the off-diagonal part; then $I - A(1) = (I - \Lambda)(I - \tilde{B})$ with $\tilde{B} = (I - \Lambda)^{-1}N$ strictly triangular in a topological order (`longrun_factor`, `longrun_inv`). Under acyclicity the AR polynomial factorises into univariate $\phi_j(z) = 1 - \sum_\tau (A_\tau)_{jj} z^\tau$; stability, $\phi_j(0) = 1$ and the intermediate value theorem give $\phi_j(1) = 1 - \lambda_j > 0$ (`selfloop_sum_lt_one`). (c) Stationarity. (d) For a linear SEM $X = BX + e$ with noise variances ω, $R_i^2 = 1 - 1/(\Sigma_{ii}(\Sigma^{-1})_{ii})$ and $(\Sigma^{-1})_{ii} = 1/\omega_i + \sum_{k\in\mathrm{ch}(i)} B_{ki}^2/\omega_k$. In an out-star with root 0 and leaves k, $1 - R_0^2 = 1/(1 + \sum_k r_k)$ and $1 - R_k^2 = 1/(1 + r_k)$ with $r_k = \omega_0 B_{k0}^2/\omega_k > 0$, so with ≥ 2 leaves the root has strictly larger R^2 than every leaf; the in-star is symmetric. Strict inequalities in the limit SEM of (b), plus (a) and continuity of $S \mapsto R^2$ at positive-definite S, give the claim for all large D. Under subsampling every contemporaneous statistic is a function of $\Gamma(0)$. □

The R^2 statement for stars strengthens the three-node analysis in Herman et al. (2025, App. D), whose qualitative message (that R^2 reflects connectivity, with an asymmetry between colliders and confounders) we credit. Numerical observations (exact population moments, not proved): mean block-mean R^2-sortability over 150 in-trees moves from 0.853 ($D = 1$) to 0.834 ($D = 4$–8) and 0.880 ($D = 256$), so it is not monotone in D; on out-stars subsampled data are R^2-sorted at 0.842 on average while block means give 0.546 ($D = 4$), 0.101 ($D = 16$) and 0 in the limit.

D. Proofs for Section 5

Notation. $\hat{C}_{ab} = \frac{1}{n}\sum_t (X_a(t) - \bar{X}_a)(X_b(t) - \bar{X}_b)$, $\hat{r}_{ab} = \hat{C}_{ab}/\sqrt{\hat{C}_{aa}\hat{C}_{bb}}$. Constants: B_{ab} as in the main text; $D_\gamma = \max_a 2\sum_h \gamma_{aa}(h)^2$; $\kappa_1 = \max_a \sum_h |\gamma_{aa}(h)|$; $\kappa_2 = \max_{a\neq b}(\sum_h |\gamma_{aa}(h)| + \sum_h |\gamma_{ab}(h)|)$. For i.i.d. sampling, $B_{ab} = 1 + \rho_{ab}^2$, $D_\gamma = 2$, $\kappa_1 = 1$.

Lemma 16 (Gaussian chaos tail)**.** *Let $g \sim N(0, I_N)$, A symmetric, $S = g^\top Ag - \mathrm{tr}A$. For $x > 0$, $P(S \geq 2\|A\|_F\sqrt{x} + 2\|A\|_{op}x) \leq e^{-x}$, and the same for $-S$.*

Lean: algebraic steps `phi_le`, `chernoff_logmgf_le`, `chernoff_choice`, `sqrt_two_v_x`; the probability statement is not machine-checked.

Proof. Diagonalise; S has the law of $\sum_i \lambda_i(g_i^2 - 1)$. For $2s\lambda < 1$, $\log Ee^{s\lambda(g^2-1)} = \varphi(s\lambda)$ with $\varphi(u) = -\frac{1}{2}\log(1 - 2u) - u = \sum_{k\geq 2}(2u)^k/(2k) \leq u^2/(1 - 2|u|)$ for $|u| < 1/2$. With $c = 2\|A\|_{op}$, $v = 2\|A\|_F^2$ and $0 \leq s < 1/c$, $\log Ee^{sS} \leq vs^2/(2(1 - cs))$. Markov's inequality with $t = \sqrt{2vx} + cx$, $q = \sqrt{2x/v}$ and $s = q/(1 + cq)$ gives exponent exactly x. This is the Gaussian case of Hsu et al. (2012, Prop. 1), who derive it from Laurent & Massart (2000); the proof here is self-contained. □

Lemma 17 (exact variance)**.** *With $\tilde{\gamma}_{ab} = \frac{1}{n}\sum_t X_a(t)X_b(t) = g^\top Ag$ for $A = C^{1/2}MC^{1/2}$, $M = \frac{1}{2n}\left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$,*

$$2n\|A\|_F^2 = \sum_{|h|<n}\left(1 - \tfrac{|h|}{n}\right)\left[\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)\right] \leq B_{ab},$$

$\|A\|_{op} \leq \kappa_2/(2n)$; on the diagonal $2n\|A\|_F^2 \leq D_\gamma$ and $\|A\|_{op} \leq \kappa_1/n$; and $\bar{X}_a \sim N(0, \bar{\sigma}_a^2)$ with $\bar{\sigma}_a^2 \leq \kappa_1/n$.

Lean: trace and Toeplitz algebra `quad_M`, `trace_MC`, `lemma12_trace`, `lemma12_fejer`, `lemma12_le_B`, `lemma12_diag_le_D`; the step $\|C^{1/2}MC^{1/2}\|_F^2 = \mathrm{tr}(MCMC)$ and the operator-norm bounds are not machine-checked.

Proof. $\|A\|_F^2 = \mathrm{tr}(MCMC)$ and, with Toeplitz blocks $(C_{ab})_{st} = \gamma_{ab}(s - t)$, $4n^2\mathrm{tr}(MCMC) = 2\sum_{s,t}[\gamma_{aa}(s - t)\gamma_{bb}(s - t) + \gamma_{ab}(s - t)\gamma_{ab}(t - s)]$ (using that γ_{aa}, γ_{bb} are even); count pairs with $s - t = h$. $\|A\|_{op} \leq \|C\|_{op}\|M\|_{op}$ and $\|C\|_{op}$ is at most the largest absolute row sum. □

Lemma 18 (deviation event)**.** *Let $\varepsilon_{ab} = \sqrt{2B_{ab}x/n} + \kappa_2 x/n + 2\kappa_1 y/n$ and $\varepsilon_d = \sqrt{2D_\gamma y/n} + 4\kappa_1 y/n$. With probability at least $1 - d(d-1)e^{-x} - 4de^{-y}$, $|\hat{C}_{ab} - \rho_{ab}| \leq \varepsilon_{ab}$ for all $a \neq b$ and $|\hat{C}_{aa} - 1| \leq \varepsilon_d$ for all a.*

Lean: deterministic steps `centred_cov_eq`, `centred_dev_offdiag`, `centred_dev_diag`, `lemma13_eps`, `lemma13_count`.

Proof. $\hat{C}_{ab} = \tilde{\gamma}_{ab} - \bar{X}_a\bar{X}_b$. Apply Lemmas 16 and 17 to each off-diagonal and diagonal $\tilde{\gamma}$, bound $|\bar{X}_a| \leq \sqrt{2\kappa_1 y/n}$ by the Gaussian Chernoff bound, and take a union bound. □

Lemma 19 (correlation perturbation). *If $a', b' \in [1-\varepsilon_d, 1+\varepsilon_d]$, $\varepsilon_d < 1$ and $|c'-\rho| \le \varepsilon$, then $|c'/\sqrt{a'b'} - \rho| \le (\varepsilon + |\rho|\varepsilon_d)/(1-\varepsilon_d)$.*

Lean: `corr_perturb` *(the hypothesis $|\rho| \le 1$ of our earlier statement is not needed).*

Proof. $|c'/\sqrt{a'b'} - \rho| \le |c'-\rho|/\sqrt{a'b'} + |\rho||1/\sqrt{a'b'} - 1|$, with $1/\sqrt{a'b'} \le 1/(1-\varepsilon_d)$ and $|1/\sqrt{a'b'} - 1| \le \varepsilon_d/(1-\varepsilon_d)$. □

Proof of Theorem 9. Set $\delta_{ab} = (\varepsilon_{ab} + |\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$. On the event of Lemma 18, Lemma 19 gives $|\hat{r}_{ab} - \rho_{ab}| \le \delta_{ab}$, so non-relatives satisfy $|\hat{r}| \le \tau$ and relatives $|\hat{r}| > \tau$ (`decide_eq_target`, `thm15_det`). Equal declared sets give equal counts and hence $\hat{s} = s_\rho^{\mathrm{pop}}$ (`sOf_cntDecl_congr`). For a VAR observed as a long stationary series, s_ρ^{pop} is the stationary value of Theorem 5 on non-self-loop edges (Proposition 7). Explicit form: with $y = x + \log(4/(d-1))_+$ (`explicit_y`) and a common threshold $\tau < \rho_{\min}$, $P(\hat{R} \ne R) \le 2d(d-1)\exp\big(-n\min(\tau, \rho_{\min}-\tau)^2/(2(\sqrt{B}+\rho_{\max}\sqrt{D_\gamma})^2)(1+o(1))\big)$. □

Argument for Proposition 10 (asymptotic). By Isserlis' theorem, $n\,\mathrm{Cov}(\tilde{\gamma}_{ab}, \tilde{\gamma}_{cd}) \to V(ab,cd) = \sum_h [\gamma_{ac}(h)\gamma_{bd}(h) + \gamma_{ad}(h)\gamma_{bc}(h)]$ (as in Lemma 17); centring is $O(1/n)$. Assuming the CLT for quadratic forms, the delta method with gradient $(-\rho/2, -\rho/2, 1)$ gives $\sqrt{n}(\hat{r} - \rho) \Rightarrow N(0, W)$, $W = V(ab,ab) - \rho[V(ab,aa) + V(ab,bb)] + \frac{\rho^2}{4}[V(aa,aa) + 2V(aa,bb) + V(bb,bb)]$, which equals B_{ab} for an independent pair. Then $\sqrt{n}\,\operatorname{atanh}\hat{r} \Rightarrow N(0, B_{ab})$ under the null and $P(|\operatorname{atanh}\hat{r}| > z_c/\sqrt{n-3}) \to 2\bar{\Phi}(z_c/\sqrt{B_{ab}})$. □

Proof of Proposition 11. $\mathrm{Cov}(X_u, X_v) = \gamma_{uv}^Y(0) + b_u b_v$ and $\gamma_{uv}^Y = 0$ for non-relatives (Proposition 7); divide by the standard deviations (`prop21_corr_std`). For relatives the sum is nonzero except on a Lebesgue-null set. All counts equal d, so every edge scores $1/2$ (`sOf_saturated`). The rate: apply Theorem 9 with target "all pairs" and $x = y = cn$, c small enough that $\delta \le \kappa_{\min}/3$ and $\tau_n \le \kappa_{\min}/3$ eventually; $|\hat{s} - 1/2| \le 1/2$ always (`abs_sOf_sub_half_le`). The onset: by the asymptotic normality of $\operatorname{atanh}\hat{r}$, the declaration probability crosses $1/2$ where $\tanh(z_c/\sqrt{n-3}) = |\kappa|$, i.e. at $n_{1/2}$ (`ncrit_eq_iff`). □

Proof of Proposition 12. $\mathrm{Cov}(X_u, Z) = b_u\sqrt{r}$ and $\mathrm{Var}Z = 1$; the partial covariance is $\gamma_{uv}^Y + (1-r)b_u b_v$ and the partial variance $\mathrm{Var}Y_u + (1-r)b_u^2$; divide and standardise (`prop31`, `prop31_std`; positive $\mathrm{Var}Y_u$ is required). At $r = 1$, $\pi = \rho^Y$ (`cor32`). For non-relatives $|\pi_{uv}| = \sqrt{\ell_u \ell_v}\sqrt{f_u f_v}$ with $f_w(r) = (1-r)/(1-r\ell_w)$ and $f_w'(r) = (\ell_w - 1)/(1-r\ell_w)^2 < 0$ (`absPi_strictAntiOn`, `absPi_eq_zero_iff`; both need $b_u b_v \ne 0$). Bounds $(1-r)\sqrt{\ell_u\ell_v} \le |\pi| \le (1-r)\sqrt{\ell_u\ell_v}/(1-\ell_{\max})$ (`absPi_lower`, `absPi_upper`). r_{crit} solves $(1-r)\ell/(1-r\ell) = \tau'$ (`rcrit_iff`, `rcrit_example`). The finite-sample bracket uses the Lipschitz bound $|g(a',b',c') - g(a,b,c)| \le L(q)\max(|a'-a|, |b'-b|, |c'-c|)$ for $g(a,b,c) = (a-bc)/\sqrt{(1-b^2)(1-c^2)}$, $|b|, |c| \le q < 1$, $L(q) = 1/(1-q^2) + 4q/(1-q^2)^2$ (`pc_lipschitz`, `thm35_bracket`), combined with Lemma 18 for (X, Z). Multi-filter driver: for $D_v(t) = b_v C(t - k_v)$ with C an AR(1) with coefficient ϕ and $Z = C(t)$,

$$\pi_{uv} = \frac{b_u b_v(\phi^{|k_u - k_v|} - \phi^{k_u + k_v})}{\sqrt{(s_u^2 - b_u^2\phi^{2k_u})(s_v^2 - b_v^2\phi^{2k_v})}},$$

nonzero for $k_u, k_v > 0$, $\phi \in (0,1)$, $b_u b_v \ne 0$ (`prop36_ar1`, `prop36_ar1_ne_zero`); in general $\pi_{uv} \propto \mathrm{Cov}(D_u - P_Z D_u, D_v - P_Z D_v)$ vanishes for all non-relative pairs iff the residual driver components are uncorrelated, which holds for a rank-one driver with $Z = F$ (`prop36_pcov`, `prop36_zero_iff`, `prop36_rank_one`). A proxy that contains summary relatives of u, v adds the terms $-c_Z(b_u q_v + b_v q_u) - q_u q_v$ to the numerator (`prop37_pcov`, `prop37_bound`). Triad identity: $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$ and, for $r > 0$, $\ell_u = \rho_{uZ}^2/r$ (`prop38_ratio`, `prop38_loading`). □

E. Simulation Checks of F1–F3

All with fixed seeds; full tables are in the supplementary material. **F1** (5 processes: in-forests with $d = 5$ and $d = 10$, self-loop persistence $\lambda \in \{0.5, 0.9, 0.97\}$, and a near-white-noise case; $n = 200$–$20{,}000$; 1,000 replications per cell; nominal, oracle-effective and fixed thresholds). The bound was never violated in 105 cells. It is loose: with a fixed threshold $\rho_{\min}/2$, it reaches 0.9 at about 3–13 times the n at which the simulated probability does, and much later with Bonferroni thresholds, where the Chernoff step loses the Mills-ratio factor. The empirical $n\,\mathrm{Var}(\hat{r})/W$ lies in $[0.93, 1.11]$. With the nominal test, $P(\hat{R} = R)$ plateaus at 0.94, 0.22, 0.03 and 0.27, as Proposition 10 predicts (predicted vs simulated false-positive rates, e.g. 0.33 vs 0.32–0.33 at $\lambda = 0.9$). **F2** ($\ell \in \{0.02, 0.1, 0.3, 0.64\}$): the asymptotic detection curve matches the simulated fraction of declared non-relatives within 0.02 for $n \ge 200$; predicted $n_{1/2} = 1{,}079$ for $\ell = 0.1$,

simulated 0.479 at $n = 1{,}000$. **F3**: predicted $n_{\text{crit}} = 472$ ($\ell = 0.64$, $r = 0.9$) gives a simulated declared fraction 0.495 at $n = 471$; over 38 cells the asymptotic prediction is within 0.03. With travel-time lags $(0, 1, 2, 4, 8)$, $\phi = 0.9$, $r = 1$, the declared fraction still rises to 0.63 at $n = 10{,}000$, as the multi-filter part of Proposition 12 predicts.

F. Lean Formalisation: Map and Findings

Lean 4.35.0 and Mathlib v4.35.0. The build log of a clean rebuild, the `#print axioms` output for each of the 116 main theorems (all `[propext, Classical.choice, Quot.sound]`) and a grep showing no `sorry`/`admit`/`axiom` are in the supplementary material.

statement	Lean names	status
Lemmas 13 and 14	`unrolled_reach_iff`, `unrolled_reach_iff_dist`, `roots_unrolled_iff`, `desc_root_unrolled_iff`, `isRel_unrolled_iff`	checked
Theorem 1	`edge_score`, `edge_score_dist`, `strictCond_iff_dist`	checked
Proposition 4	`rho_unrolled_eq`, `mT_le_iff`, `rho_unrolled_large_t`	checked
Theorem 2	`sRho_unrolled_exact`, `sRho_unrolled_bounds`, `tendsto_sRho_unrolled`, `tendsto_sRho_GT`	checked
Corollary 3	`tendsto_half_iff`, `crossEdges_eq_empty_tfae`, `sRho_unrolled_ge_of_cross`, `reisach_theorem4`, `limitValue_dag`	checked
Theorem 5	`rhoBurn_eq`, `sBurn_eq`, `rhoZ_eq`, `sZ_eq`, `sOf_window`, `relSet_eq_iff_src`, `mem_upEdges_iff`	checked
Corollary 6	`sBurn_eq_half_iff`, `sBurn_eq_half_iff_common_anc`, `sZ_eq_half_iff`, `upEdges_eq_empty_tfae`, `stat_half_of_strong`, `stat_half_of_outForest`, `inForest_isRel_iff`, `inForest_up_iff`, `statValue_inForest`, `statValue_inForest_gt_iff`	checked
Table 1	`collider_limit`, `fork_limit`, `chain3_limit`, `collider_stat`, `fork_stat`, `chain3_stat`	checked
self-loop remark	`edgeLoops_limit`, `edgeNoLoops_sRho`	checked
Proposition 7	–	paper only
Theorem 8(a)	`fejer_identity` (counting identity)	partial
Theorem 8(b)	`longrun_factor`, `longrun_inv`, `selfloop_sum_lt_one`	partial
Theorem 8(c,d)	–	paper only
Lemma 16	`phi_le`, `chernoff_logmgf_le`, `chernoff_choice`, `sqrt_two_v_x`	algebra only
Lemma 17	`quad_M`, `trace_MC`, `lemma12_trace`, `lemma12_fejer`, `avg_toeplitz_le_tsum`, `lemma12_le_B`, `lemma12_diag`, `lemma12_diag_le_D`	algebra only
Lemma 18	`centred_cov_eq`, `centred_dev_offdiag`, `centred_dev_diag`, `lemma13_eps`, `lemma13_count`	deterministic part
Lemma 19	`corr_perturb`	checked
Theorem 9	`decide_eq_target`, `decide_eq_support`, `thm15_det`, `explicit_y`, `sOf_cntDecl_congr`	deterministic part
Proposition 10	–	asymptotic
Proposition 11	`prop21_corr`, `prop21_corr_std`, `sOf_saturated`, `sOf_const`, `abs_sOf_sub_half_le`, `thm22_const`, `ncrit_iff`, `ncrit_eq_iff`	deterministic part
Proposition 12	`pcorr_eq_pcov`, `pcorr_eq_corr_resid`, `prop31`, `prop31_std`, `cor32`, `prop33`, `prop33_abs`, `absPi_lower`, `absPi_upper`, `absPi_le_kappa`, `absPi_lt_kappa`, `absPi_eq_zero_iff`, `absPi_strictAntiOn`, `absPi_equal`, `rcrit_iff`, `rcrit_eq_iff`, `rcrit_example`, `pc_lipschitz`, `thm35_bracket`, `prop36_pcov`, `prop36_zero_iff`, `prop36_rank_one`, `prop36_ar1_pcov`, `prop36_ar1_pvar`, `prop36_ar1`, `prop36_ar1_ne_zero`, `prop37_pcov`, `prop37_bound`, `prop38_ratio`, `prop38_loading`	population part checked

Modelling choices. Summary graphs are relations on a finite type with a self-loop hypothesis; anc and desc are reflexive-transitive closures ("a path exists iff a walk exists" is the one convention adopted without proof); G_T with $T = n + 1$ slices

lives on $V \times \mathrm{Fin}(n+1)$; source SCCs are represented by their member nodes (equality of SCC sets as equality of node unions is elementary and not formalised). In the finite-sample files, zero-mean variables are vectors of an arbitrary real inner-product space and lag-0 covariance is the inner product; independence enters only as vanishing inner products.

Findings. No statement was false. Formalisation added four missing nondegeneracy hypotheses to our earlier finite-sample statements: (i) the strict claims of Proposition 12 need $b_u b_v \neq 0$; (ii) $\ell_u = \rho_{uZ}^2 / r$ needs $r > 0$; (iii) nonvanishing of the multi-filter residual needs $b_u b_v \neq 0$; (iv) the partial correlation needs positive partial variances ($\mathrm{Var} Y_u > 0$ suffices). It also showed that one hypothesis of Proposition 4 and one of Lemma 19 are redundant, and that the in-forest statement needs acyclicity, not only out-degree at most one.

G. Real-Data Details

CausalRivers (Stein et al., 2025). Study 1 (P1–P3): the benchmark's connected-subgraph sampler, seeds 7000–7029, East Germany and Bavaria, $d \in \{5, 10\}$, 120 subgraphs, 791 induced edges; block means at 12 h and 24 h; nominal Fisher-z. Study 2 (P4–P6): seeds 8000–8029, 120 subgraphs, 1,376 non-relative pairs; proxy Z is the mean of standardised stations of the region outside the subgraph and outside 2 undirected river hops (429–615 stations); primary test on partial correlations with a residual Bartlett n_{eff} ($K = 100$ steps at 12 h, 50 at 24 h); median n_{eff} 139/113 (raw) and 231/211 (residual). Before conditioning, 0.98 (nominal) and 0.87 (effective) of non-relative pairs tested dependent. Median proxy contamination by river relatives was 2.6% (maximum 72%), a case of the contaminated-proxy term in Proposition 12. Post hoc and without verdict: conditioning on $k = 1, 3, 5, 10$ principal components of the leave-out stations gave dependent fractions $0.745, 0.605, 0.576, 0.491$ (12 h), with accuracy 0.580 and Spearman 0.16 at $k = 10$.

Causal Chambers (Gamella et al., 2025). Candidate selection (recorded before analysis) compared Causal Chambers, CausalRivers with measured precipitation (rejected: the driver reaches each gauge through its own catchment filter, so by Proposition 12 a scalar proxy cannot remove it), a semi-synthetic cross-sectional manufacturing benchmark (rejected: not real, not time series) and fMRI with physiological regressors (rejected: no ground truth). Wind tunnel: runs `actuators_random_walk_2-16` (run 1 excluded before analysis: truncated); 16 random variables, 26 edges, 63 non-relative pairs per run; block means over 10 rows, first differences, standardisation, $n = 999$; effective-n Fisher-z with Bartlett $K = 20$. Light tunnel: `actuators_white`, 20 variables, 39 edges, ten segments of 2,000 steps, levels. Predicted values come from the package's published ground-truth graph with constant parameters removed. Secondary analyses (no verdicts): nominal n gives slightly more false positives (0.048), as Proposition 10 predicts; levels instead of increments gave 0.054, contradicting our own secondary prediction that levels of random walks would produce many spurious dependences; without block means ($D = 1$) relative recovery drops to 0.18 and Spearman to 0.10. In the light tunnel, polariser angles act on two sensors through Malus' law, a nonlinearity with near-zero linear correlation; as pre-registered, these relative pairs were never detected, without affecting orientation.

H. Pre-Registration Timeline and Deviations

Times are commit timestamps relative to the first pre-registration commit (T_0); all events occurred on one day. The plan 4 commit was made about 3 min 36 s before its results commit, which includes a run of about two minutes; as before, the process is short and agent-driven. Commit hashes are withheld for anonymity and will be provided with the de-anonymised version. Commit timestamps are set by the committing client; the hosting service's push records are the stronger evidence and can be supplied to the area chair on request.

time	event
$T_0 + 0{:}00$	plan 1 committed: targets T1–T3, predictions P1–P3, kill rule
$T_0 + 0{:}14$	theory of Sections 3 and 4, enumeration checks, operationalisation of P1–P3 (before data)
$T_0 + 0{:}17$	study 1 results (P2, P3 fail; kill rule triggered)
$T_0 + 1{:}05$	plan 2 committed: F1–F3 targets, Lean target, predictions P4–P6, kill rule
$T_0 + 1{:}18$–$1{:}31$	Lean formalisation of Section 3 and algebraic parts of Section 4
$T_0 + 1{:}38$	finite-sample theory, simulations, operationalisation of P4–P6 (before data)
$T_0 + 1{:}44$	study 2 results (P4, P6 fail; kill rule triggered)
$T_0 + 4{:}58$	plan 3 committed: candidate scoping, graph-only predictions, P7–P9 (before any dependence statistic)
$T_0 + 5{:}02$	study 3 results (P7, P8 pass; P9 fails; replication passes)
$T_0 + 5{:}14$–$5{:}24$	Lean formalisation of the finite-sample lemmas; errata
$T_0 + 5{:}45$	manuscript draft committed (with placeholders for E1/E2)
$T_0 + 5{:}50$	plan 4 committed: E1 (test of the theorem against eight naive predictors), E2 (lagged relatives), theory addendum A1–A3 and B1–B3, graph-only predictions, design check; development runs on the study-3 data only; no dependence statistic of any held-out dataset computed
$T_0 + 5{:}54$	E1/E2 results committed as pre-registered (E1 passes 8/8 on the primary graph; Q7, Q8, Q10 pass, Q9 fails); out-of-sample F checks; run logs

Deviations (all recorded at the time). (1) Plan 1 said "theorem-predicted s_ρ" without naming the version; the stationary version was fixed in writing before data were loaded (the finite version would predict $> 1/2$ for every subgraph, including chains). (2) Self-loops are excluded from the predicted average because the estimator cannot orient them; this does not change which side of $1/2$ the prediction is on. (3) Only 6 of 120 study-1 subgraphs lacked a confluence; the plan set no minimum size for that group, and its trivial pass is not interpreted. (4) Plan 2 did not say which test defines "dependent" in P4; the effective-n test was fixed as primary before data, and the nominal test fails P4 as well. (5) In study 3, a planned triad-reliability check was not applicable (the measured driver reaches only four mutually related barometers) and is not interpreted. (6) The P9 failure is close to the threshold and its CI contains 0.3; as pre-registered, the point estimate decides. (7) Plan 4 (E1/E2): no threshold, unit, seed, K, L, D, graph or node set changed after the results were seen. A stray cache file committed with plan 4 was removed in the results commit (no content change). The F-check tolerance of ± 0.03 is the one fixed in plan 4; the citris "predicted accuracy" is undefined (no full-set edge of G_{lin} is predicted 1). The constant-$1/2$ comparison is post hoc. The choice of G_{lin} as the primary graph was pre-registered, together with the secondary analysis on the published graph G_D, which is the one on which the theorem fails against the sensor indicator on strictly ordered edges.

I. Experiment E1: Discriminating Test Against Naive Predictors

Status and provenance. Pre-registered in plan 4 (Appendix H); the plan was committed before any dependence statistic of any held-out dataset was computed. Done before it: file hashes, column and row counts, regeneration of the `citris_1` input protocol from its published generator (it matches the recorded inputs exactly), graph-only predictions, and development runs on the study-3 data only. All numbers below come from the committed result files of the experiment folder (`results4.json`, `summary4.txt`); none is estimated or rounded beyond the printed digits.

Why `citris_1`. Every published Causal Chambers graph is bipartite except the pressure-control configuration, whose only experiment varies no independent input, so we rejected it. The light-tunnel `citris_1` series (100,000 steps about 1.8 s apart) is generated by a time-series structural model among the actuators: red is a reflected random walk, green moves halfway towards red and blue a quarter of the way towards green (each plus a uniform step), pol_1 is a random walk and pol_2 moves towards or away from pol_1 according to $\text{sign}(\text{blue} - \text{red})$. With the sensors this is a chain of length 3, red $\to$ green $\to$ blue $\to$ sensors. The secondary data are snapshot (i.i.d.) structural models among the actuators (`buchholz_1_obs`; `scm_{1,2,4,5}_reference`, including a chain and a collider) and the wind-tunnel regime-jump runs (bipartite; disagreements arise only from latent-node ties).

Linear-zero lemma (the refinement). Write $\mathcal{C}$ for the colour block (red, green, blue, ir_1, vis_1, ir_2, vis_2, ir_3, vis_3, current) and $\mathcal{Q}$ for the polariser block (pol_1, pol_2, angle_1, angle_2). *B1 (proved for the actuators).* Conditionally on the flag and the colour paths, the law of the polariser paths is invariant under negation, because every update (uniform draw on a symmetric interval, move towards or away with a sign fixed by the colours, reflection at ± 90, rounding to 0.1) commutes with negation. *B2 (assumed).* The first-two-layer sensors depend only on the colours and independent noise; ir_3 and vis_3 depend on the polarisers only through an even function of $\theta_1 - \theta_2$ (Malus' law with aligned zero marks); each angle sensor is affine in its polariser plus independent noise. This is a physical assumption that the design documents do not let us check. *B3.* Under

B1 and B2 every variable of $\mathcal{C}$ has zero cross-covariance, at every pair of times, with every variable of $\mathcal{Q}$ (a product of an even and an odd function of the polariser path has conditional mean zero). The six real edges red $\to$ pol_2, blue $\to$ pol_2 and $\text{pol}_k \to \text{ir}_3$, vis_3 therefore carry no linear dependence. The relatives that a correlation-based estimator targets are those of the *linear-dependence graph* G_{lin} (the design graph G_D without these six edges), which has two weak components, each with a single source (red; pol_1). By Corollary 6, every edge inside a block has predicted value $1/2$ for every node set; the six cross-block edges are compared by block size: red $\to$ pol_2 and blue $\to$ pol_2 are predicted 0 (anti-sorted) and $\text{pol}_k \to \text{ir}_3$, vis_3 are predicted 1 for the full set. Faithfulness inside the blocks is assumed, as in Proposition 7. A zero-mark offset or a non-affine angle sensor would break B2 and move the prediction towards G_D.

Metric and tests. For each edge instance (the full node set plus 20 subgraphs per unit) the observed value is $o \in \{0, \frac{1}{2}, 1\}$ from the estimated counts, and the loss is $|o - p|$ for the predicted value p. For each competitor take the instances where it disagrees with the theorem; $\Delta = \text{loss(theory)} - \text{loss(competitor)}$, pooled, with a unit-level exact sign-flip test (1,024 flips for 10 units) and Holm correction over eight competitors, and a unit-bootstrap CI. E1a (prerequisites): non-relatives declared < 0.5 and relatives detected ≥ 0.8. E1b: $\Delta \leq -0.10$ and Holm $p \leq 0.05$ for all eight. E1c: the same restricted to instances whose own pair is declared dependent ($\Delta < 0$). Separation was to be claimed iff E1a, E1b and E1c all pass on `citris_1`. Kill rules: K1 (E1a fails: no claim), K2 (a competitor is significantly better: no claim against it), K3 (E1b passes and E1c fails: attributed to power-induced ties). None was triggered on the primary analysis.

Table 4. E1 primary, `citris_1`, graph G_{lin}, lag 0, 1,662 instances. Δ [95% CI]; all rows: 10/10 units, Holm $p = 0.0078$. "Exact": share of instances whose predicted value equals the observed one. "Spearman": agreement of the mean predicted edge value with observed accuracy over 200 subgraphs.

predictor	Δ [CI]	Δ, detected edges	exact	Spearman
theorem	—	—	0.992	0.962
sensor indicator	-0.502 [-0.507, -0.496]	-0.500	0.203	0.408
ancestors, full graph	-0.515 [-0.519, -0.510]	-0.500	0.147	0.334
ancestors, subgraph	-0.509 [-0.513, -0.505]	-0.500	0.120	0.121
in-degree, full graph	-0.515 [-0.520, -0.511]	-0.500	0.180	0.349
in-degree, subgraph	-0.510 [-0.514, -0.505]	-0.500	0.143	0.105
depth, full graph	-0.515 [-0.519, -0.510]	-0.500	0.147	0.334
depth, subgraph	-0.510 [-0.514, -0.505]	-0.500	0.114	0.113
every edge 1	-0.533 [-0.536, -0.530]	-0.500	0.115	undefined

Observed values given the theorem's prediction: predicted $1/2$ (1,365 instances) observed exactly $1/2$ in 99.2%; predicted 1 (187 instances, polariser $\to$ ir_3, vis_3) observed 1 in 99.5%; predicted 0 (110 instances, red and blue $\to$ pol_2) observed 0 in 99.1%. The detected-edge subset has 1,338 instances, all observed exactly $1/2$. Prerequisites: non-relatives declared 0.000 [0, 0] (40 pairs per unit), relatives detected 0.969 [0.945, 0.992] (51 pairs per unit).

Strict subset and the power safeguard. On the 297 instances where the theorem predicts 0 or 1 it beats all eight ($\Delta = -0.539$ for the sensor indicator, -0.948 for ancestors and depth on the full graph, -0.991 for "every edge 1"; Holm $p = 0.0078$, 10/10 units). There, an estimator without power would favour the competitors (the "all counts tied" baseline is $+0.443$ to 0), so this part cannot come from power-induced ties.

Published graph G_D (secondary). Applying the theorem to the unmodified design graph, relatives detection is 0.659 [0.643, 0.675] (75 relative pairs per unit) and non-relatives declared 0.000. Overall the theorem still beats all eight ($\Delta = -0.323$ for the sensor indicator; -0.424 for ancestors, depth and "every edge 1"; -0.420 and -0.417 for in-degree; 10/10 units, Holm $p = 0.0078$), and its exact-match rate is 0.714. But on the 442 edges where G_D predicts strict order (1), the theorem matches 0.222 and the sensor indicator 0.319; the sensor indicator is significantly better ($\Delta = +0.453$ [0.410, 0.492], theorem better in 0/10 units, reverse Holm $p = 0.002$). A constant-$1/2$ rule (post hoc) matches 0.815 of all instances, against 0.714 for the theorem on G_D and 0.992 on G_{lin}; mean $|o - 1/2| = 0.092$ against 0.166 for the theorem on G_D. The Spearman statistic is 0.341 (theorem on G_D) against 0.408 (sensor indicator). This is why the claim is restricted to the linearly dependent relatives.

Lagged estimator on `citris_1` (secondary). With $L = 5$ and G_{lin}: detection 0.976 [0.953, 0.992], non-relatives declared 0.000, 8/8 passed ($\Delta = -0.505$ to -0.535), theorem exact match 0.998, Spearman 0.996.

On the snapshot sets, relatives detected: 0.933 (`buchholz_1_obs`), 1.000 (`scm_1`), 0.965 (`scm_2`), 1.000 (`scm_4`, `scm_5`); non-relatives declared 0.000 wherever non-relative pairs exist (`scm_4` has none). The chain (`scm_4`) gives $1/2$ on all

Table 5. E1 on i.i.d. structural-model sets (no verdict; secondary). Competitors' exact-match range over the eight predictors. Units: `scm_1` 5, the others 10 each (45 pooled).

set	instances	theorem exact	competitors exact	Δ range	result
`buchholz_1_obs`	1,712	0.791	0.328–0.375	-0.499 to -0.496	pass 8/8
`scm_1_reference`	867	1.000	0.314–0.392	-0.500	not met[†]
`scm_2_reference`	1,650	0.812	0.324–0.408	-0.443 to -0.426	pass 8/8
`scm_4_reference` (chain)	2,267	1.000	0.000–0.088	-0.500	pass 8/8
`scm_5_reference` (collider)	2,238	1.000	0.464–0.495	-0.500	pass 8/8
pooled (45 units)	8,734	0.923	0.283–0.322	-0.489 to -0.487	pass 8/8, Holm $p = 8 \times 10^{-5}$

[†]Five units: the smallest attainable Holm-adjusted p is $8/32 = 0.25$.

2,267 instances (observed exactly $1/2$ every time); the collider gives 1 on edges into the collider and its sensors and $1/2$ elsewhere (all 2,238 instances exact). Pooled, the theorem's strict predictions number 2,768 (observed value 1 on 2,403, $1/2$ on 349, 0 on 16: exact 0.868); on them only the sensor indicator disagrees often (390 instances, $\Delta = -0.5$, 35/35 units) and the other competitors coincide with the theorem. The pooled Spearman is 0.913 for the theorem and -0.138 for the sensor indicator. Post hoc, a constant-$1/2$ rule matches 0.688 of the pooled instances against 0.923.

Wind-tunnel regimes (tertiary, 64 units, no verdict). The same test passes against all eight in each configuration. $D = 10$, $L = 0$: non-relatives declared 0.014, relatives detected 0.592, $\Delta = -0.203$ $[-0.227, -0.179]$ in 63/64 units, exact match 0.539 against 0.432; detected-edge subset $\Delta = -0.169$. $D = 1$, $L = 30$: 0.003, 0.398, $\Delta = -0.225$ $[-0.250, -0.197]$ in 62/64 units, exact 0.443 against 0.353; detected-edge subset $\Delta = -0.134$. $D = 1$, $L = 0$: $\Delta = -0.218$ (64/64 units), exact 0.411 against 0.326; $D = 10$, $L = 3$: $\Delta = -0.218$, exact 0.533 against 0.411. Detection of 0.33–0.59 means that part of the advantage is power-induced, as plan 4 stated in advance. Post hoc, the constant-$1/2$ rule has mean absolute error 0.311 against 0.363 for the theorem at $D = 1$, $L = 30$ (and 0.313 against 0.294 at $D = 10$, $L = 0$), and matches 0.378 (0.375) of the instances against 0.443 (0.539).

What E1 does and does not show. It shows that, on one multi-layer light-tunnel series, the estimated relatives counts are tied along the whole chain red $\to$ green $\to$ blue $\to$ sensors and between the polariser block and the angle sensors, and strictly ordered (including against the causal direction) on red, blue $\to \mathrm{pol}_2$ — and that the source-component rule of Theorem 5 and Corollary 6 predicts this while eight naive rules do not. It does not show that the theorem predicts relatives under the published graph; the faithfulness of the graph to linear dependence fails here by design. Most of the separation is in the ties, which naive rules cannot express. The evidence is one device in a favourable regime, and the wind tunnel, where detection is limited by F1, gives weaker support.

J. Experiment E2: Lagged Relatives

Theory (plan 4, addendum A). For $L \in \{0, 1, \dots\}$ let the lag-L relatives of u be u and every v with $\gamma_{uv}(h) \neq 0$ for some $|h| \leq L$. *A1 (proved).* Under the independent-innovations assumption behind Proposition 7, $\mathrm{rel}_0 \subseteq \mathrm{rel}_L \subseteq \mathrm{rel}_\infty \subseteq \mathrm{rel}_S$; for a VAR model all are equal for almost every parameter value; and Theorem 5, Corollary 6 hold verbatim with "relatives" read as rel_L under lag-L faithfulness. Lag-0 faithfulness implies lag-L faithfulness; the converse fails (a white-noise source s with an edge $s \to j$ and no self-loop at s has $\gamma_{sj}(0) = 0 \neq \gamma_{js}(1)$). *Estimator.* Declare u, v relatives iff $\max_{|h| \leq L} |\operatorname{atanh} \hat{r}_{uv}(h)| \sqrt{\hat{n}_{\mathrm{eff},uv}(h) - 3} > z_c(L)$, with $\hat{n}_{\mathrm{eff},uv}(h) = (n - |h|)/\hat{B}_{uv}$ and $z_c(L) = \Phi^{-1}(1 - \alpha/(2m(2L + 1)))$, a Bonferroni correction over the m pairs and the $2L + 1$ lags; $L = 0$ is exactly the study-3 test. *A2 (proved for finite n).* For independent stationary Gaussian a, b the null variance of the lag-h cross-covariance has the same bound at every h, so one Bartlett factor serves all lags. *A3 (deterministic part proved, constants sketched).* The recovery bound of F1 holds with $\rho_{\min}$ replaced by $\rho^*_{\min,L} = \min_{\text{relative pairs}} \max_{|h| \leq L} |\rho_{ab}(h)| \geq \rho_{\min}$ (the gain) and a factor $2L + 1$ in the failure probability and the Bonferroni count (the cost): for 105 pairs at $\alpha = 0.01$, z_c rises from 3.90 ($L = 0$) to 4.35 ($L = 3$) and 4.80 ($L = 30$). A lagged correlation must therefore exceed the lag-0 correlation by about 1.12 ($L = 3$) or 1.23 ($L = 30$) times to help.

Design. Held-out data: the wind-tunnel regime-jump runs (`regime_jumps_single`, `regime_jumps_multi`), $2 \times 32 = 64$ regimes of 10,000 steps, in the same archive as the study-3 data but never analysed in study 3. Load_in, load_out, pot_1, pot_2 follow independent mirrored random walks within a regime; the hatch and sensor parameters are constant within a regime. The first 100 rows of each regime are dropped. Graph: 20 edges without the hatch (19 predicted 1, $\mathrm{pot}_1 \to \mathrm{signal}_1$

predicted $1/2$); 54 non-relative and 51 relative pairs; 1,280 subgraphs (10 of size 5 and 10 of size 8 per unit). Primary: $D = 1$ (first differences), $L = 30$ (about 9 s) against $L = 0$; secondary: block means with $D = 10$ and $L = 3$ against $L = 0$. Success required Q7–Q10 all to pass at the primary configuration. Kill rules: Q7 fails (saturation) or Q10 fails (lags do not help detection, and the lagged estimator is not recommended). Plan 4 states that E2 is not a re-analysis of P9 and that P9's verdict is final.

Table 6. E2, 64 held-out regimes, 95% unit-bootstrap CIs. Q8: accuracy on the 1,216 instances predicted 1. Q10: relatives detected, lagged minus lag 0 (non-relatives declared difference in the last column).

configuration	Q7	Q8	Q9	Q10	non-rel. diff.
primary D=1, L=30	0.003 [0.001, 0.005] pass	0.553 [0.527, 0.579] pass	0.243 [0.193, 0.294] **fail**	$+0.069$ [0.060, 0.079] pass	$+0.001$ [-0.001, 0.002]
secondary D=10, L=3	0.011 [0.008, 0.014]	0.711 [0.692, 0.729]	0.202 [0.152, 0.253] fail	-0.007 [-0.012, -0.003] fail	-0.002 [-0.004, -0.0003]
lag 0, D=10 (replication of study 3)	0.014 [0.010, 0.018]	0.742 [0.722, 0.760]	0.202 [0.145, 0.259] fail	—	—
lag 0, D=1	0.002 [0.001, 0.004]	0.552 [0.532, 0.570]	0.120 [0.066, 0.176] fail	—	—

Detection of the 51 relative pairs: $D = 1$: 0.328 [0.312, 0.345] at lag 0 and 0.398 [0.383, 0.412] with $L = 30$; $D = 10$: 0.592 [0.584, 0.599] and 0.585 [0.576, 0.592] with $L = 3$ (lags reduce detection: the Bonferroni cost exceeds the gain). On `citris_1` with $L = 5$ against $L = 0$: 0.976 against 0.969, difference $+0.008$ [0, 0.024], so Q10 fails there; detection is already nearly saturated at lag 0. **Verdict.** Q9 fails at the primary configuration, so E2 does not succeed; no kill rule is triggered (Q7 and Q10 pass). Lags raise detection at fine resolution without adding false positives, as A1 and A3 predict. The study-3 estimator (lag 0) rerun on these held-out regimes fails the graded criterion again (0.202 at $D = 10$, 0.120 at $D = 1$): the P9-type failure repeats on held-out data. P9's original verdict (0.267, failed) stands; E2 is not presented as rescuing it. The development check of plan 4 showed that, on the study-3 data with the same estimator, drawing different subgraph seeds moves the P9 statistic from 0.267 to 0.338. We report this as variability of the statistic (about 0.07 from the subgraph draw alone); it is a development run, not a verdict, and it does not change the P9 outcome.

Table 7. Detection rate per edge over 64 regimes: share of regimes in which the pair was declared dependent.

edge	D=1, L=0	D=1, L=30	D=10, L=0
load_in → rpm_in	0.859	0.875	1.000
load_in → rpm_out	0.000	0.172	0.094
load_in → current_in	1.000	1.000	1.000
load_in → current_out	0.000	0.000	0.047
load_out → rpm_in	0.000	0.016	0.000
load_out → rpm_out	0.844	0.969	1.000
load_out → current_out	1.000	1.000	1.000
load_out → current_in	0.000	0.000	0.000
load_in → pressure_intake	0.000	0.094	0.953
load_out → pressure_intake	0.000	0.031	0.906
load_in → pressure_upwind	0.562	0.984	1.000
load_out → pressure_upwind	0.016	0.812	1.000
load_in → pressure_downwind	0.734	0.953	1.000
load_out → pressure_downwind	0.750	1.000	1.000
$\text{pot}_1 \to \text{signal}_1$	0.359	0.156	0.969
$\text{pot}_1 \to \text{signal}_2$	0.266	0.109	0.969
$\text{pot}_2 \to \text{signal}_2$	0.000	0.000	0.266
pot_1, load_in, load_out → mic	0.000	0.000	0.016 (pot_1), 0.000

Lags recover the pressure edges at fine resolution (load_in → upwind pressure $0.562 \to 0.984$, load_out → upwind pressure $0.016 \to 0.812$) but lose the pot edges to the larger threshold ($0.359 \to 0.156$ and $0.266 \to 0.109$); every microphone edge, $\text{pot}_2 \to \text{signal}_2$ and the cross-current edges stay undetected. Even at its best ($D = 10$, lag 0) wind-tunnel detection (0.592) is far below that of the light tunnel, so lags are not the main cause of the F1 shortfall.

All six checks pass. With lags the observed detection sits at or slightly below $\bar{p}_{\max}$, consistent with strongly correlated lag statistics; the independence formula overstates both detection and false positives, as the theory said it would. For `citris_1` the predicted accuracy is undefined because no full-set edge of G_{lin} is predicted 1 inside a block.

Table 8. Out-of-sample F-theory checks: the first half of each unit predicts the second half. Predicted detection is a bracket $[\bar{p}_{\max}, \bar{p}_{\mathrm{ind}}]$ when lags are used. Pass if the observed value lies within 0.03 of the bracket (criterion fixed in plan 4).

configuration	predicted detection	observed	non-rel. declared, pred. / obs.	accuracy, pred. / obs.
wind D=1, L=0	0.250	0.242	0.007 / 0.003	0.573 / 0.590
wind D=1, L=30	[0.344, 0.426]	0.332	[0.021, 0.074] / 0.003	0.504 / 0.506
wind D=10, L=0	0.548	0.533	0.014 / 0.008	0.649 / 0.632
wind D=10, L=3	[0.536, 0.554]	0.524	[0.016, 0.027] / 0.008	0.628 / 0.618
`citris_1`, L=0	0.935	0.945	0.007 / 0.000	undefined
`citris_1`, L=5	[0.940, 0.988]	0.945	[0.010, 0.045] / 0.000	undefined